

Predictive Analytics for Sustainable Fisheries
A MACHINE LEARNING APPROACH TO TRIP PROFITABILITY, COST AND
SAFETY IN SRI LANKAN OFFSHORE FISHING

Group Assignment - Semester II

ST 3011 - Statistical Programming

Department of Statistics - Faculty of Science, University of Colombo

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Abstract

This project uses data from over 20,000 Sri Lankan offshore fishing trips from 2022 to 2025 period to help fishermen and boat owners make better decisions before leaving the port. As we know Sri Lanka's offshore fishing industry faces huge uncertainty due to fluctuating fuel costs, unpredictable catch volumes and some safety risk. Fishermen and fleet managers lack data-driven tools to predict trip profitability and costs or identify high-risk voyages before departure. This project aims to address these inefficiencies by developing statistical models that predict economic outcomes and classify operational risks. Under 20,000 observation and by looking at more than 30 different variables, this study investigates how digital tools can predict a trip's success.

The data preprocessing consisted cleaning operational metrics, feature engineering, as well as imputation of missing numerical values using a Multivariate Imputation by Chained Equations (MICE) Algorithm. Then, Tree-based models like Random Forest and boosting methods (XGBoost, LightGBM) are used because they effectively handle the high multicollinearity and non-linear patterns found in fishing trip data by focusing on feature importance rather than strict variable independence. We specifically implement CatBoost to process categorical vessel and port data natively, which prevents information loss during encoding to provide more accurate predictions for trip profitability and safety compared to standard linear models.

Finally, the end goal is to reduce financial uncertainty and improve maritime safety by making these advanced models accessible through an interactive website.

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Introduction

Local fishermen in Sri Lanka are struggling with rapidly changing fuel prices and many problems caused by the different tragic events country faced such as COVID-19 pandemic, the 2022 financial crisis, and recent floods. These events have made every trip a huge financial risk. If a boat returns with a small catch, the owner faces massive losses of five figures or more. Even worse, inaccurate weather forecasts can lead to tragic deaths at sea. Because of these dangers, fishermen desperately need a recommendation system. This analysis would help them check potential costs, profits and safety risks before they leave the port, ensuring they don't risk their lives or their savings on a failing trip.

This project introduces a predictive framework to shift the industry from reactive to proactive decision making. By analyzing over 20,000 historical records of Sri Lankan fishing trips, we examine how pre-departure signals such as vessel specifications, fuel inputs and sea conditions can explain variations in cost/profitability and safety. This analysis designed to provide real-time support so fishermen can minimize costs and maximize safety before even leaving the port.

Description of The Question We Are Going to Answer

The primary research question guiding this data analysis is “Can we use data and weather forecasts to predict if a fishing trip will be profitable and safe before the boat even leaves the port?” The goal is to put all this information into an easy-to-use website that helps the fishing community make better, safer, and more profitable decisions. In order to address this question, we look at three distinct paths of investigation,

1. **Productivity Prediction (Net Profit Prediction):** The goal is to predict the expected net profit (Revenue - Cost) of a trip based on operational inputs.
2. **High/ Low Productive Trip Classification:** The goal is to flag trips that are likely to be “High Productive” or not.
3. **Cost Optimization Research (Trip Cost Prediction):** The goal is to predict the total operational cost of a fishing trip before it departs. This helps boat owners plan budgets.
4. **Marine Risk Assessment (Risk/Safety Classification):** The goal is to warn fishermen about high-risk weather to keep them safe.

Description of The Data Set

“Sri Lankan Offshore Trip Data (2022 - 25)” dataset focuses on the offshore and deep-sea fishing sector of Sri Lanka. This sector is critical to the national economy and supporting higher number of active fishermen and contributing significantly to the national fish harvest.

The dataset contains records of approximately 20,000 offshore fishing trips. There are total 39 variables (both numerical and categorical) and those can be divided in to three sub categories such as;

- I. Operational Metrics: Vessel type (e.g., IMUL, NTRB), boat dimensions, environmental factors (wind speed, air visibility, wave height etc.), number of days etc.
- II. Input Costs: Fuel consumed (liters), crew size, ice and gear usage etc.
- III. Output Metrics: Total catch weight (kg), species breakdown (tuna, billfish, etc.), and gross revenue (LKR) etc.

Column	Description	Column	Description
year	Year of trip record	fuel_liters	Total fuel consumption (liters)
trip_id	Unique identifier for the fishing trip	fuel_price_per_liter_LKR	Fuel price per liter (from CEYPETCO scraping)
vessel_choice	Whether vessel is defined by type or class	fuel_cost_LKR	Total fuel expenditure (LKR)
boat_type_or_class	Boat type (IMUL/IDAY/OFRP/MTRB/etc) or length class (28_34ft/35_39ft/etc)	ice_cost_LKR	Cost of ice used for catch preservation
boat_length_ft	Length of vessel in feet (only for class-based vessels)	food_cost_LKR	Food cost for crew based on crew size and days
engine_hp	Engine horsepower	medicine_cost_LKR	Medical expenses adjusted for rough weather
port_lat	Latitude of departure port in Sri Lanka	gear_cost_LKR	Cost of fishing gear used during the trip
port_lon	Longitude of departure port in Sri Lanka	crew_wage_LKR	Total crew wages based on days * fixed wage rate
zone_lat	Latitude of fishing zone generated around port	variable_cost_LKR	Daily operational cost (only for class vessels)
zone_lon	Longitude of fishing zone generated around port	license_fee_LKR	Standard license fee added for every trip
distance_km	Haversine distance between port and offshore zone	total_cost_LKR	Total operational cost of the trip
trip_hours	Total hours spent for full trip (travel + fishing)	revenue_LKR	Revenue obtained from fish catch after tax deduction
no_days	Total number of offshore days allowed by vessel type/class	gross_profit_LKR	Profit after subtracting total_cost from revenue
crew_size	Number of crew members onboard	trip_type	Shallow (<10 km) or deep (8%~10 km) fishing classification
month	Month of departure	safety	Weather safety category (safe or unsafe)
wind_kph	Wind speed during trip (kph)	Yellowfin_Tuna	Catch weight of Yellowfin Tuna (kg)
wave_m	Wave height during trip (meters)	Skipjack	Catch weight of Skipjack Tuna (kg)
precip_mm	Estimated rainfall (mm)	Marlin	Catch weight of Marlin (kg)
visibility_km	Visibility distance (km)	Other	Catch weight of other species (kg)
departure_hour	Departure hour of the trip		

Data Pre-processing, Feature Engineering and Imputation Strategy

The “trip_id” column was removed since unique identifiers typically does not carry predictive power for machine learning models and that deletion reduces noise.

The “year” variable was transformed into “tripAge” by subtracting the recorded year from the current year. This new variable is often more useful for identifying trends related to the age of the trip.

Using the **geopy** library, departure port’s latitude and longitude coordinates were converted into particular city of the port of Sri Lanka. This transforms raw spatial data into categorical location data (departurePortLocation).

To handle broad marine coordinates in the sea, **K-Means Clustering** was used to group trips into 15 distinct “Fishing Zones”. By searching the domain information, we found that the fisheries sector is primarily organized around 15 fishery zones in sea range in Sri Lanka. This simplifies complex coordinated into a single categorical variable. Which allows the model to understand the specific “neighborhoods” where fishing occurs.

A numerical variable “boat_length_ft” exhibited high percentage of missing values as 49.36%. To address this while preventing data loss, Multivariate Imputation by Chained Equations (MICE) was applied. This method effectively predicted and imputed these missing values by utilizing relationships found within the existing data structure.

New variables like “fuelCostLKR” and “netProfit(LKR)” were engineered by performing arithmetic on raw columns (fuel_liters * fuel_price_per_liter_LKR and revenue_LKR - total_cost_LKR). This ensures data integrity by using calculated values rather than potentially inaccurate pre-existing columns. Then, used raw columns were dropped to avoid redundancy.

Several columns were renamed to be more descriptive. This is a vital preprocessing step for interpretability and understanding while coding. This ensures that the model output clearly communicates exactly what each metric represents.

```
columnMapping = {
    'trip_hours': 'totalHoursSpent(travelAndFishing)',
    'no_days': 'allowedOffshoreDays',
    'month': 'MonthOfDeparture',
    'precip_mm': 'estimatedRainfall(mm)',
    'visibility_km': 'visibilityDistance(km)',
    'safety': 'safetyCategory',
    'Yellowfin_Tuna': 'catchweightOfYellowfinTuna(kg)',
    'Skipjack': 'catchweightOfSkipjack(kg)',
    'Marlin': 'catchweightOfMarlin(kg)',
    'Other': 'catchweightOfOtherFish(kg)'
}
```

```
masterFishingData.rename(columns=columnMapping, inplace=True)
```

Under each different objectives, data subsets were chosen (feature selection) from the master data set and done different data preprocess steps accordingly.

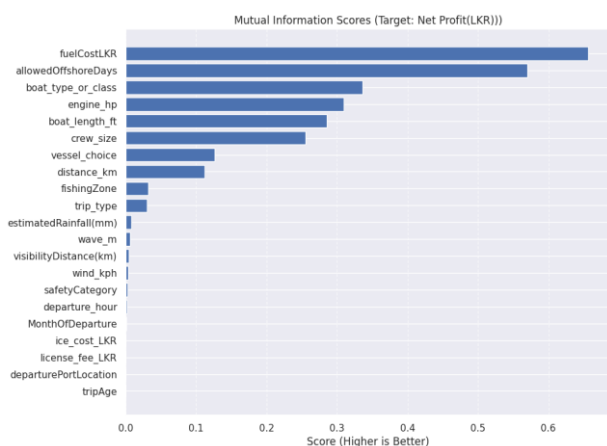
Objective I: Productivity Prediction (Net Profit Prediction)

Removing Redundant & Direct Future Result Related Variables

Since the predictive model is practically applicable for a fisherman still at the port, several variables must be removed to prevent Data Leakage. Data leakage occurs when a model is trained on "future" information. Hence, the **future variables** such as “final revenue”, “different costs” and “specific catch weights” were removed. Including these outcomes, along with variable expenses like “paid crew wages” (which are often share-based) or the “actual hours spent at sea”, would allow the model to cheat by seeing the results it is supposed to predict. Additionally, redundant variables like food, medicine, and gear costs are removed because they are mathematically tied and related to existing features like “crew size” and “boat type”.

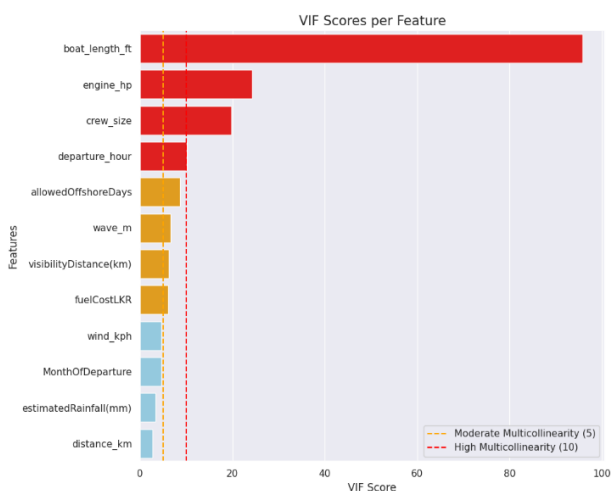
Feature Selection (Using MI Scores)

Features with a Mutual Information (MI) score is equal to zero, such as “tripAge” “ice_cost_LKR”, “license_fee_LKR”, and “departurePortLocation”, were removed because they share no statistical dependency with the target variable (Independent). Meanwhile, this shows that fuelCostLKR and allowedOffshoreDays are the most critical factors for predicting net profit.



Check Multicollinearity

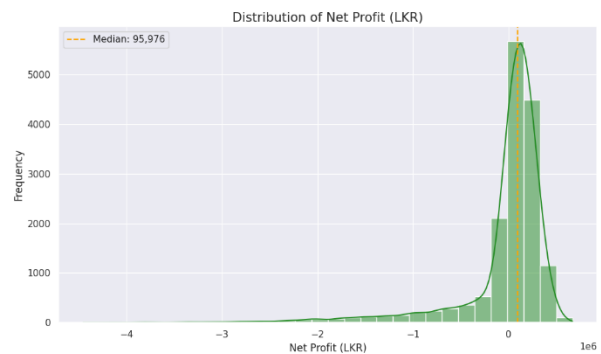
VIF results show that boat_length_ft has a critically high score near 100, indicating severe multicollinearity with other features. Additionally, engine_hp, crew_size, and departure_hour all exceed the high multicollinearity threshold of 10, which can destabilize model training and make it difficult to determine the individual impact of these variables. In contrast, environmental factors like wind speed and distance remain in the safe zone with low VIF scores.



Exploratory Data Analysis (Target: Net Profit)

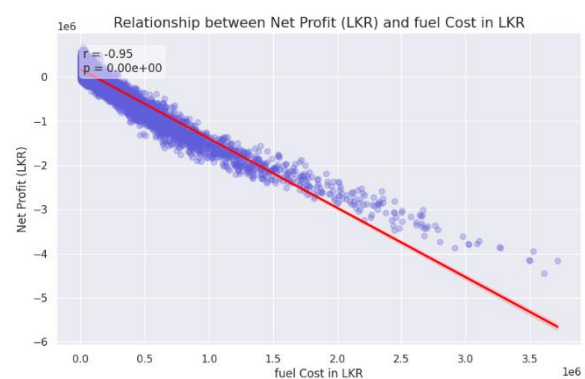
Distribution of Net Profit (LKR)

The distribution of Net Profit is strongly left-skewed, with the median of 95,976 LKR. While most trips result in a modest profit, a significant "long tail" extends into the negatives, showing some trips losing over 4,000,000 LKR. The deep losses in the tail may represent "failed trips" caused by mechanical breakdowns, poor weather or returning with an empty hold despite high fuel expenditures.



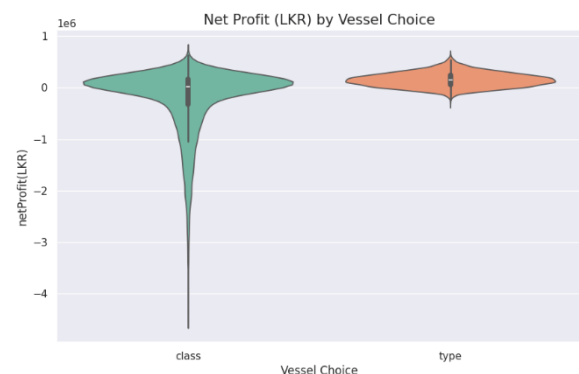
Relationship between Net Profit (LKR) and fuel Cost in LKR

There is a near-perfect negative correlation ($r = -0.95$). As fuel costs increase, net profit decreases linearly into deep negative values. This means that fuel is a 'major cost' that dominates the profit equation. There is a dense concentration of data points where fuel costs are low since vessels consume a baseline amount of fuel and yield a modest catch.



Net Profit (LKR) by Vessel Choice

The plot reveals that the 'type' vessel is far more stable, with a small spread and the distribution concentrated around positive profit. In contrast, the 'class' vessel shows a much wider spread and a massive downward tail into extreme negative values (exceeding -4,000,000 LKR). This happens due to the 'class' vessels likely have significantly higher operational overheads such as massive fuel requirements or larger crews leading to high cost.



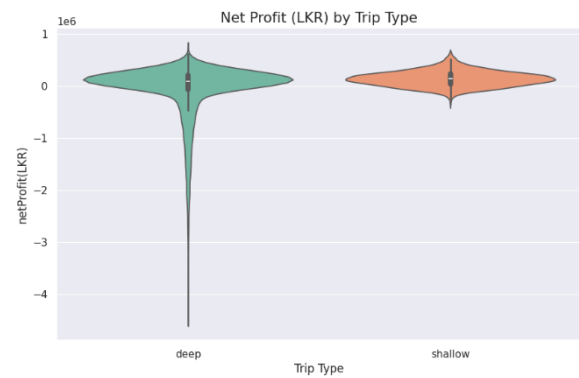
Net Profit (LKR) by Safety Category

The plot shows that while both "safe" and "unsafe" categories have similar median profits while the 'unsafe' category has a comparatively long 'tail' of extreme losses, reaching down to nearly 5,000,000 LKR since operating in 'unsafe' trips introduces extreme financial volatility when things go wrong at sea.



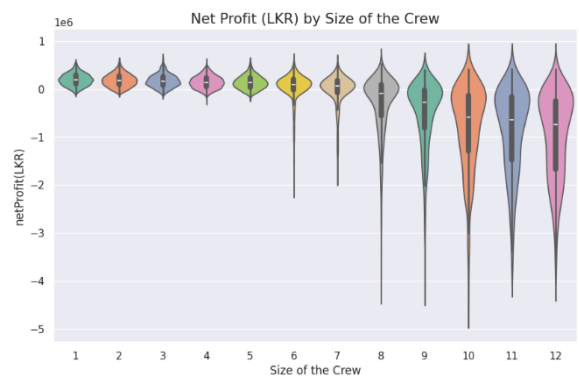
Net Profit (LKR) by Trip Type

The plot shows that deep sea trips have a massive downward tail, meaning they can lose up to 5,000,000 LKR on a single journey. Meanwhile, shallow water trips stay close to the middle, meaning their profits and losses are much smaller and more predictable. The reason for this gap is operating cost. If a deep sea boat travels far and finds no fish, it still has to pay for a huge amount of fuel.



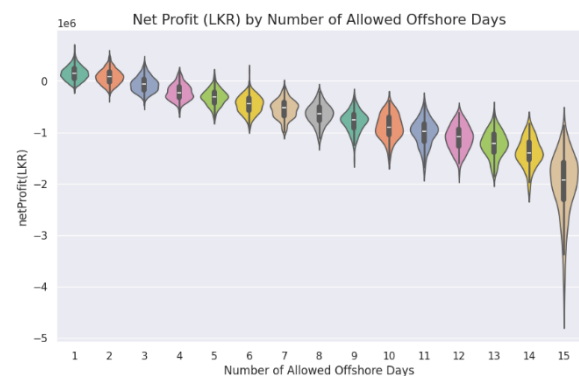
Net Profit (LKR) by Size of the Crew

This violin plot showing that while smaller crews (1-5 members) maintain stable, positive profit margins, larger crews (8-12 members) experience significant volatility. As crew size increases, the median profit trends downward and the risk of substantial financial loss grows. Which is evidenced by the increasingly long downward 'tails' and wider interquartile ranges for larger groups.



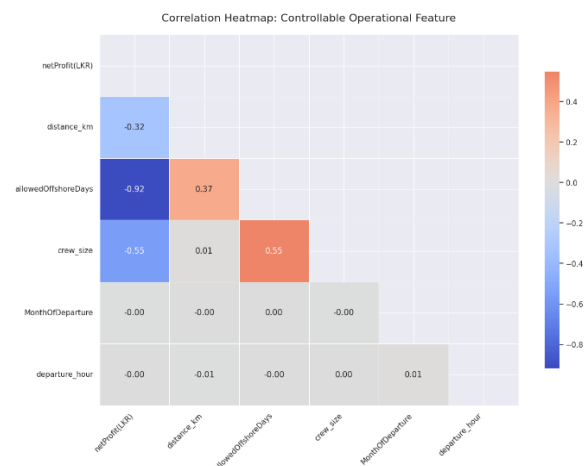
Net Profit (LKR) by Number of Allowed Offshore Days

The plot shows a strong negative relationship between the number of allowed offshore days and net profit. Profitability is highest and most stable during short trips of 1 to 2 days, after which median profits consistently trend into negative values. As the duration extends toward 15 days, the data shows a dramatic increase in financial risk which is characterized by significantly larger losses and increased spread.



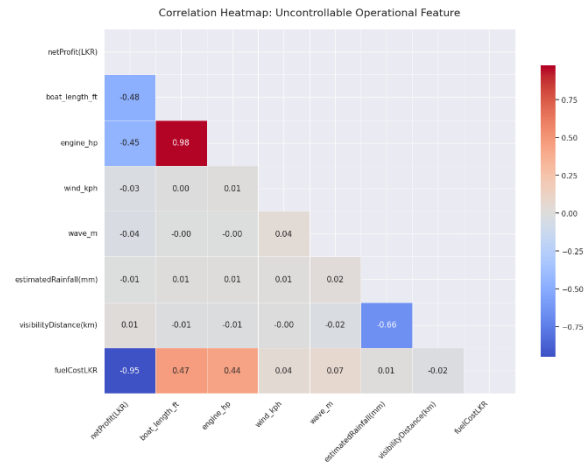
Correlation Heatmap: Controllable Operational Feature

Distance between sea location and the port, allowed offshore days & crew size show negative correlations with net profit. While departure month does not show any correlation with net profit in this dataset. Further, multicollinearity seems to be present in the dataset and it should be addressed in any future analysis.



Correlation Heatmap: Uncontrollable Operational Feature

Both boat length (-0.48) and enginehp (-0.45) demonstrate moderate negative correlations with net profit. This means that larger and more powerful vessels in this fleet tend to be less profitable due to the higher overhead and fuel consumption associated with their size. A nearly perfect positive correlation of 0.98 exists between boat length and enginehp. This must be addressed in future predictive modeling to avoid skewed results.



Inferential Statistics

To assess the median differences in net profit across various operational factors, conducted a Mann-Whitney U test. This non-parametric approach was specifically chosen because the target variable, Net Profit (LKR) is not normally distributed and heavily skew, making traditional parametric tests inappropriate. Here the results

i. Checking median differences in net profit across between deep and shallow trip types
Mann-Whitney U Statistic: 15531578.00

P-value: 0.0000

Conclusion: Reject the null hypothesis. There is a statistically significant difference in the medians of netProfit(LKR) between deep and shallow trip types.

ii. Checking median differences in net profit across between different vessel choice

Mann-Whitney U Statistic: 18041423.00

P-value: 0.0000

Conclusion: Reject the null hypothesis. There is a statistically significant difference in the medians of netProfit(LKR) between 'class' and 'type' vessel choices.

iii. Checking median differences in net profit across 'safe' and 'unsafe' safety categories
Mann-Whitney U Statistic: 29536478.00

P-value: 0.0203

Conclusion: Reject the null hypothesis. There is a statistically significant difference in the medians of netProfit(LKR) between 'safe' and 'unsafe' safety categories.

Data Preprocessing

Feature Scaling & Encoding: Continuous variables were scaled (Fitted only for train data and transformed to both train, test data). Although, tree-based and boosting models generally do not require feature scaling, this step ensured consistency across all the models. Then, Categorical predictors were encoded using dummy encoding. This preserved category level information while making the data suitable for machine learning algorithms. (But, Catboost model does not need manual encoding. Hence, we ran the Catboost model before the categorical encoding)

Hyperparameter Tuning

A hyperparameter tuning is done to identify the most effective configurations for each algorithm. For CatBoost, utilized **Optuna** which is a complex Bayesian optimization framework that efficiently explores the parameter space. For the remaining models LightGBM, DecisionTreeRegressor, Random Forest and XGBoost, implemented **GridSearchCV** with **5-fold cross-validation** to ensure that the models generalize well across different subsets of the train data and reducing the risk of overfitting.

Model Building and Results

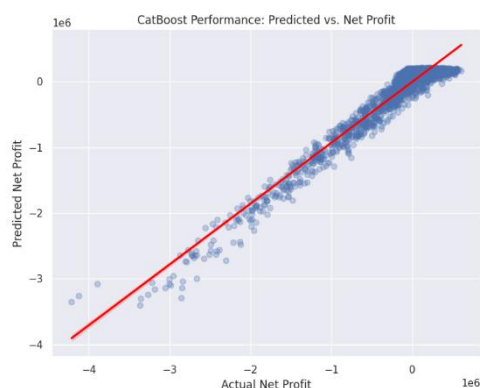
Model	Train R2	Test R2	Train RMSE	Test RMSE	Absolute Difference R2
Random Forest	0.94405	0.932581	118814.4597	134597.6399	0.011469
Catboost	0.929882	0.931191	133010.2306	135977.5961	0.001309
lightgbmRegressor	0.933624	0.930972	129412.8549	136193.6866	0.002652
XGBoost	0.928873	0.930683	133963.9508	136479.2234	0.00181
DecisionTreeRegressor	0.925642	0.927427	136972.513	139646.9175	0.001785

Five different models were developed and checked for the R squared values. We choose **CatBoost** over **Random Forest**. Although Random Forest has the highest Test score, the drop from a 0.944 train score indicates it is memorizing noise (Overfitting Risk). But catboost's performance is almost identical on both train and test sets, suggesting it handles unseen data perfectly. Also, can note that the Train RMSE > Test RMSE (and the corresponding Test R^2 > Train R^2) is generally okay and indicates a highly robust model for unseen data as well.

According to the results, CatBoost has a good Test R^2 of 0.9325 meaning that it explains over 93% of the variance in Net Profit (LKR) when seeing new data. Also, CatBoost natively handles categorical features and high-dimensional data without the need for extensive manual encoding, which leads to a better performance than remaining boosting models. And its internal "Ordered Boosting" mechanism is specifically designed to reduce bias and handle high-dimensional feature sets more effectively than other models.

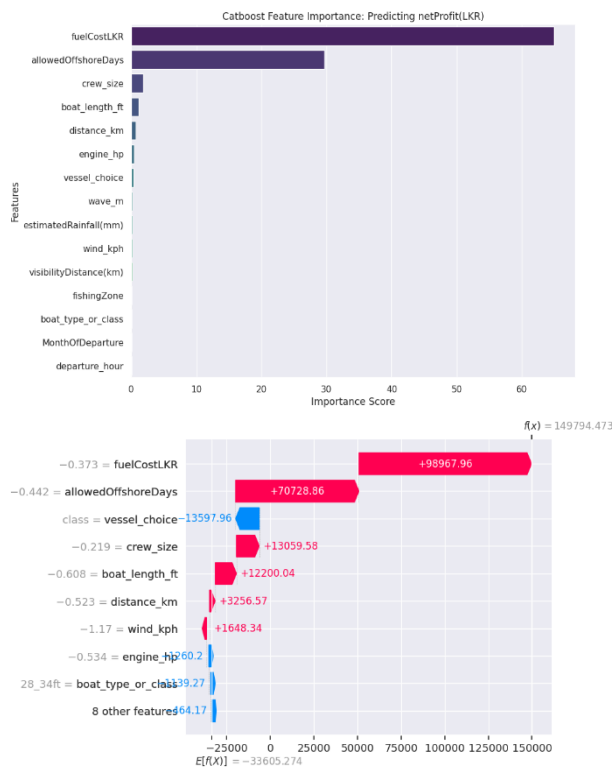
Next, XGBoost very stable and performs similarly to Catboost and LightGBM gives a good performance. However, we chose Catboost as the best choice.

Visualization of Catboost Model Performance



The Predicted vs. Actual Net Profit plot for the test data shows the model stays tight to the diagonal line across the entire range. Which suggests that model has high accuracy

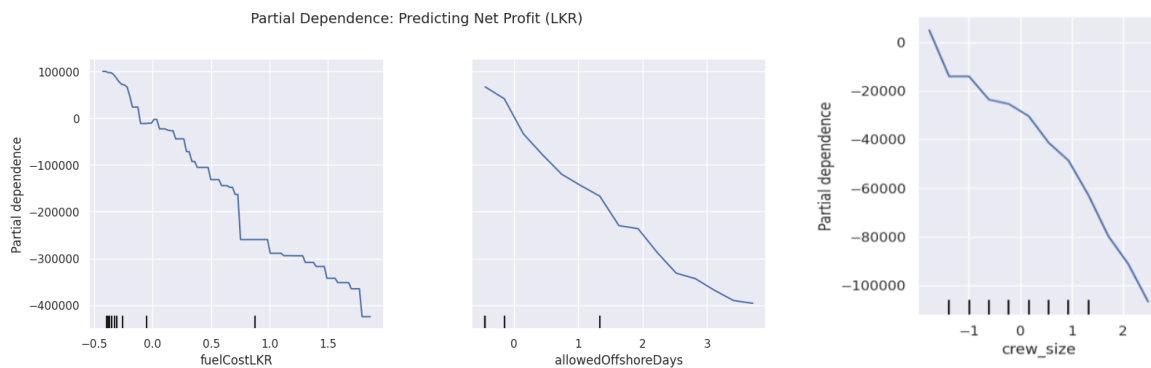
Catboost Feature Importance



We can identify Fuel Cost is the most critical feature with an importance score of approximately 65. Which is followed by Allowed Offshore Days is the second most influential factor. Features like crew_size, boat_length_ft, and distance_km have much lower importance scores (below 5), suggesting they have a smaller, though still present, impact on the final profit prediction.

The SHAP waterfall plot illustrates how specific features push the model's prediction from a base value (average) to the final predicted value for a single trip. Features like Fuel Cost Allowed Offshore Days significantly increase (red) the predicted net profit. Factors such as vessel_choice and engine_hp pull the predicted profit down(blue).

Partial Dependencies



Partial dependency plot shows both Fuel Cost and Allowed Offshore Days show a clear downward nonlinear trend. As these costs or durations increase beyond certain thresholds, the marginal net profit prediction decreases sharply. We can see specific steps in the graphs where the predicted profit drops significantly as well.

Objective II: High/ Low Productive Trip Classification:

Removing Redundant & Direct Future Result Related Variables

Just like the previous objective, the same set of variables were removed to prevent Data Leakage and the variable which store the future outcome values were removed at first.

Feature Selection (Using MI Scores)

Features with a Mutual Information (MI) score is equal to zero, such as Visibility Distance(km) and license_fee_LKR were removed because they share no statistical dependency with the target variable (Independent). Meanwhile, this shows that expected Ice Cost to be brought is the most critical factors for predicting net profit.

ice_cost_LKR	0.609464
fuelCostLKR	0.155651
allowedOffshoreDays	0.140337
boat_type_or_class	0.115888
boat_length_ft	0.115049
engine_hp	0.112996
crew_size	0.093513
vessel_choice	0.041519
distance_km	0.007928
trip_type	0.004466
wave_m	0.002731
fishingZone	0.002105
estimatedRainfall(mm)	0.001613
wind_kph	0.000474
safetyCategory	0.000290
MonthOfDeparture	0.000259
departure_hour	0.000094
departurePortLocation	0.000061
tripAge	0.000059
visibilityDistance(km)	0.000000
license_fee_LKR	0.000000

Feature Engineering

Created a new binary variable called “Productivity Status” by using the median as a threshold (since net profit target variable is skewed). It makes sure a perfectly balanced dataset where 50% of the trips are labeled "High Productive" and 50% are "Low Productive". Then, target variable was replaced by Productivity Status and dropped previous target Net Profit.

Data Preprocessing

Same feature scaling and encoding was done as previous one. Since, Catboost Classifier does not need manual encoding. Hence, we ran the Catboost Classifier model before the categorical encoding)

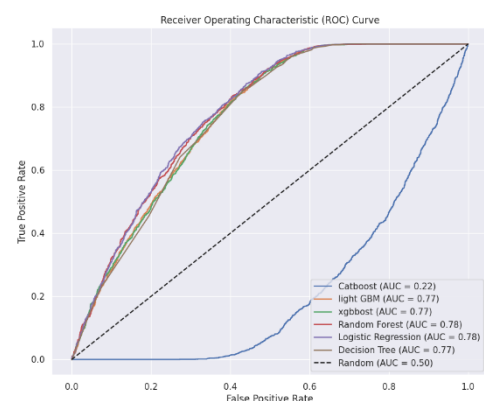
Hyperparameter Tuning

A hyperparameter tuning is done to identify the most effective configurations for each algorithm. In here we’ve used GridSearchCV with 5-fold cross-validation to ensure that the models generalize well across different subsets of the train data and reducing the risk of overfitting.

Model Building and Results

To model the ‘Productivity Status’ for this classification problem, we implemented a range of models consisting of baseline, tree-based and boosting classifiers. We utilized Logistic Regression as the primary linear baseline, tree-based analysis included Decision Trees and Random Forest and advanced boosting frameworks such as Catboost, XGBoost, and LightGBM classifier to capture complex non-linear patterns.

Classifier	Accuracy	ROC_AUC	Recall	Precision	F1
Catboost	71.875000	0.221234	0.921287	0.658295	0.767898
Logistic Regression	71.600000	0.782219	0.894554	0.661905	0.760842
Random Forest	71.475000	0.779729	0.916832	0.655575	0.764499
xgboost	71.275000	0.768492	0.870297	0.664650	0.753698
light GBM	70.850000	0.768310	0.870792	0.660285	0.751067
Decision Tree	70.725000	0.765339	0.893069	0.653860	0.754970



According to the results, Catboost classifier achieved a high accuracy but a very low ROC_AUC (0.224). An AUC below 0.50 suggests the model’s probabilities are inverted or the

model is failing to generalize properly. And logistic regression maintains a high Recall (0.904) which keeping a healthy Precision (0.655), offering a more balanced prediction for a real-world setting. After that we have the runner up Random Forest Classifier.

But I chose **Random Forest Classifier** over logistic regression since the existence of high multicollinearity in this data causes the coefficients in a Logistic Regression to become unstable. which makes the model overly sensitive to small changes in data. And unlike Logistic Regression, Random Forest can capture complex interactions rather than linear relationships.

Hence, Random Forest was selected as the final model due to its brilliant ability to handle high-dimensional feature spaces and high multicollinear data.



Objective III: Cost Optimization Research (Trip Cost Prediction):

Preparing the data

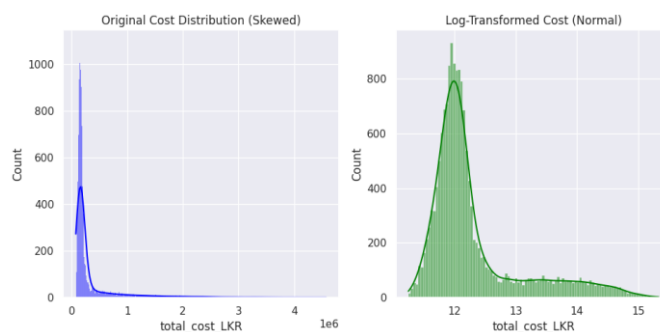
In the data preparation always, the model remained “blind” to any information gathered during the trip. This involves removing the cost components like fuel cost, food cost and catch weights which directly influence the target variable “Total Cost”, forcing the model to rely on the weather, physical conditions of the vessel and the trip planning data such as the crew size, trip type and allowed off shore days. Categorical features were transformed using One-Hot Encoding, while numerical features were inspected for missing values. To handle the significant right-skewness of the cost data, a log-transformation was applied.

Exploratory Data Analysis

Exploratory Data Analysis was conducted with the normal target values and also the log transformed values. The clearer pictures were obtained for the log transformed plots because of the high skewness in the data. Therefore, the following plots are plotted with the log transformed target variable.

Log Transforming the Price

The original cost data is extremely right-skewed, by shifting the distribution from right skewed tail towards a normal curve the transform says that the model errors are consistent across both short distance and long-distance voyages.



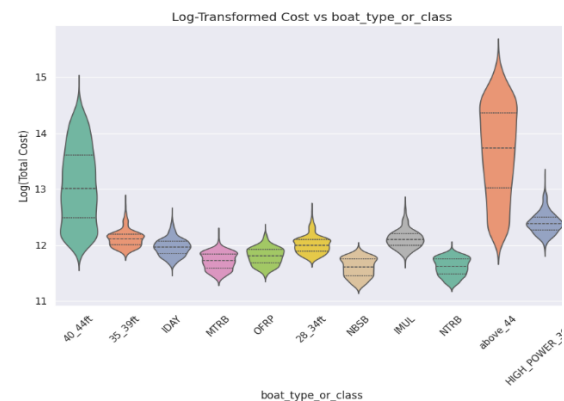
Log Transformed Cost Vs Trip Type

After log transforming the distribution become clearer, reducing skewness and making central tendencies clear for the both trip types. Overall, the plot suggest that the deeper trips are generally more expensive.



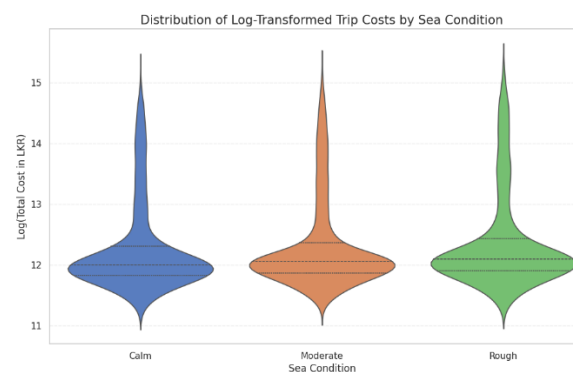
Log Transformed Cost Vs Boat Type or Class

The violin plot shows that the mid-sized categories such as 40–44 and HIGH POWER >300HP also have relatively higher costs compared to smaller classes like NTRB, NBSB, and MTBB, which cluster at lower log-cost levels. Overall, cost tends to increase with boat size/class, and variability becomes larger for higher-end boat categories.



Log Transformed TRIP Costs by Sea Condition

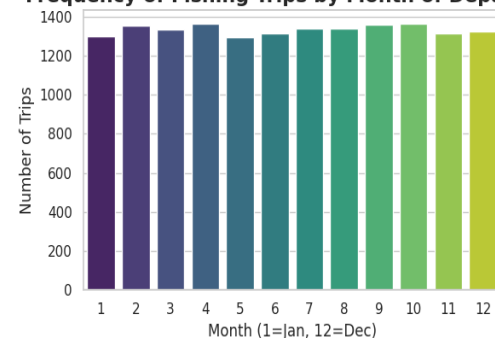
As the results in the plot which has plotted with the price itself didn't give a clear picture, here the log transformed price was used. The median costs slightly increase from calm to rough conditions indicating higher typical expenses in harsher seas. Rough conditions also show wider spread, suggesting greater variability and risk in trip costs under rough seas.



Frequency Of Fishing Trips by Month of Departure

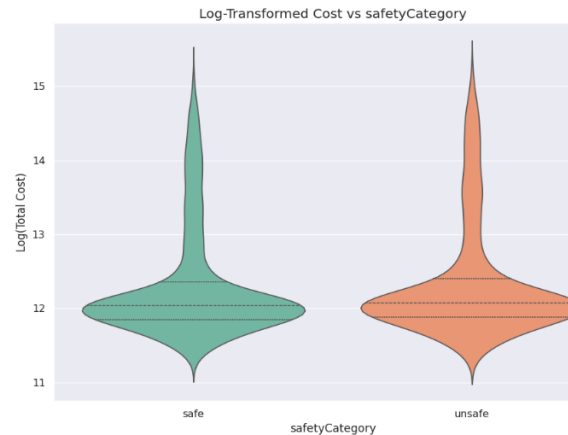
The bar chart shows that fishing trips are evenly distributed across all months, with only small fluctuations throughout the year. There are slight increases in trips during mid to late months, suggesting a seasonal peak. Overall, no none of the months dominate indicating consistent fishing activity year-round.

Frequency of Fishing Trips by Month of Departure



Log Transformed Cost Vs Safety Category

The violin plot shows that both safe and unsafe categories have similar central log-cost values, clustered around 12. The unsafe category has a wider spread and higher upper tail indicating that greater variability and more high-cost trips. This suggests that while average costs are comparable, unsafe trips are more likely to incur extreme expenses.



Multicollinearity Assessment and The Statistical Tests

Before modeling VIF values were checked to clarify whether the multicollinearity exist. Furthermore, non-parametric statistical tests were used to validate the importance of specific categories. **Mann-Whitney U Test** confirmed that there is a statistically significant difference in costs between the trips labeled as “Safe” and “Unsafe”. Additionally, a **Chi-square test of Independence** performed between weather conditions and safety categories. The resulting p-value proved that weather and safety are dependent which means the bad weather explain that it’s not safe. Following are the tables values.

- Variance Inflation Factor (VIF)	
Feature	VIF
boat_length_ft	24.217020
engine_hp	20.836657
crew_size	3.633029
allowedOffshoreDays	1.763567
distance_km	1.225762
wave_m	1.002121
wind_kph	1.002063

Mann-Whitney U statistic: 27394993.5000

P-value: 6.862e-08

Conclusion: Reject the Null Hypothesis ($p < 0.05$).

There IS a statistically significant difference in the distribution of costs between Safe and Unsafe trips.

--- Chi-Square: Link between Weather and Safety ---

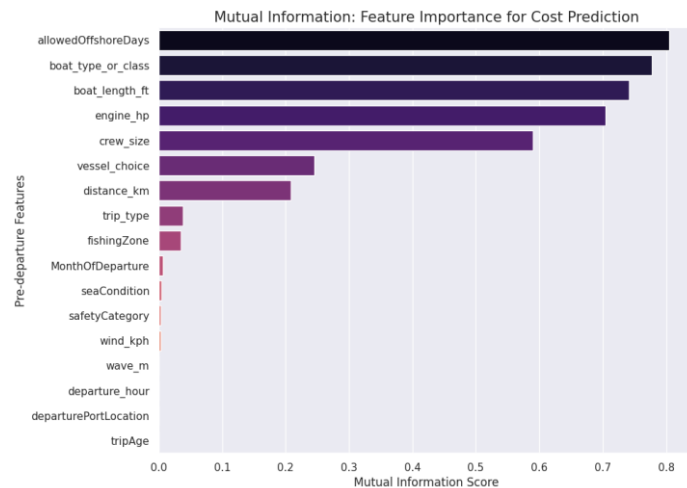
Chi-Square Statistic: 9365.6536

P-value: 0.0000

Conclusion: Weather and Safety are DEPENDENT. Rough weather is statistically linked to safety risks.

Mutual Information

The plot shows that allowed Off shore Days, boat type or class, and boat length are the most informative features for predicting cost, followed by engine hp and crew size. Mid-level contributors include vessel choice and distance(km), while variables like trip type and fishing Zone add only minor information.



Data Preprocessing

All the numerical variables were scaled during the modeling process and all the categorical variables were encoded using one-hot encoding.

Model Development and evaluation

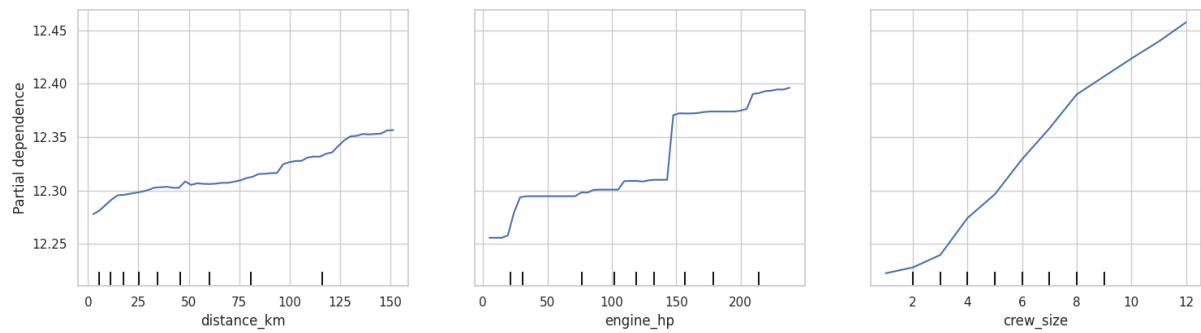
Five distinct models were developed and checked for the R squared values. At first models were fitted using the original Total Cost variable. The RMSE scores were very high due to the large scale of data. Therefore, models were again fitted using the log-transformed Total Cost variable. Then the loss functions were quit low and the R squared values were quite high. Following are the results after training the model using the log transformed target variable.

Model	Train R2	Test R2	Train MAE (LKR)	Test MAE (LKR)	Test MSE (LKR)	Test RMSE (LKR)
XGBoost	0.984247	0.979215	20223.097091	27121.307335	4.238163e+09	65101.177062
Random Forest	0.996843	0.978300	10745.076769	29132.641639	4.951544e+09	70367.209700
Ridge	0.963531	0.964419	48257.039038	49807.641487	1.446451e+10	120268.482688
Elastic Net	0.958727	0.959585	54834.954527	56399.431524	1.968309e+10	140296.441065
Lasso	0.955897	0.956720	57086.023712	58350.201481	2.067825e+10	143799.351322

It has one of the highest tests R^2 value (0.9792) and the lowest test MAE (27,121 LKR) and the lowest RMSE (65,101 LKR), meaning it explains the most variance and make the smallest prediction errors on unseen data. Random Forest also close, but overall XGBoost still performs slightly better overall.

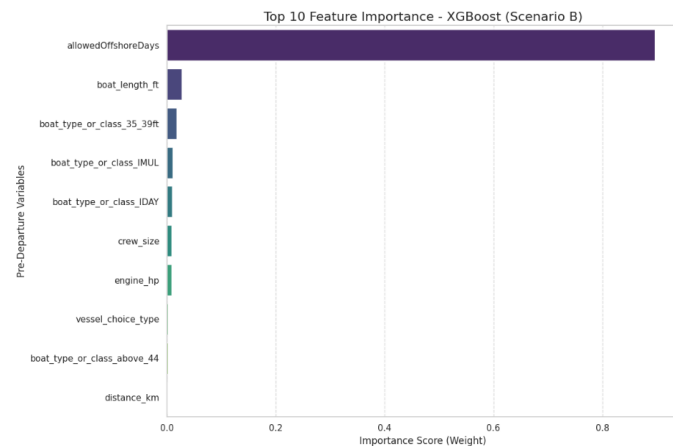
Following are the partial dependencies of the mostly influenced variables. It says when increasing the distance after keeping every variable constant, the total price is getting increased. Same scenario happens with engine_hp and crew size.

Partial Dependence of Trip Cost (Log-Scale) on Proxies



Feature Importance of XGBoost

Cost is being driven mainly by how many offshore days are allowed, with vessel characteristics and the crew size. The similar results were gained from the mutual information plot too. Therefore, the model is relying primarily on offshore permission days to predict cost, and the rest of the features play only a minor supporting role.



Conclusion

This research shows that it is possible to estimate fishing trip costs before departure with high accuracy. Among all the model tested, XGBoost performed best. The study also identified “Weather”, “Labor” and “Zone” have a major impact on costs. The results will help boat owners to decide about the crew size and the departure timing, to lower the cost which leads for mor sustainable fishing operations.

Objective IV: Marine Risk Assessment (Risk/Safety Classification)

Preparing the data

The first stage involved removing cost based and economic variables from the dataset. Variables related to fuel cost, trip expenditure, revenue, or other monetary measures were excluded because they do not have a direct causal or practical relationship with the Safety Category and retaining these variables could introduce noise and weaken the interpretability of the safety model. In the second stage, variables were further removed based on mutual information (MI) was equal to zero. This MI based filtering ensured that only predictors with meaningful explanatory power were retained, leading to a more compact, informative and generalizable feature set for subsequent modelling.

Multicollinearity Assessment

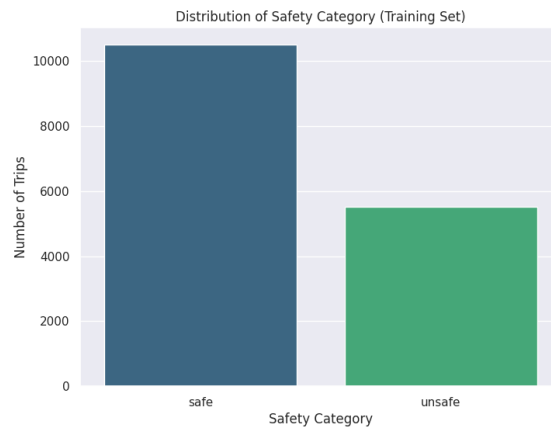
Multicollinearity among continuous predictors was systematically evaluated using Variance Inflation Factor (VIF) analysis to quantify the extent of interdependence between variables. The assessment revealed that no variables exhibited problematically high VIF values indicating that all continuous predictors could be retained without concerns regarding redundancy or

numerical instability. While tree-based algorithms are inherently robust to multicollinearity due to their recursive partitioning mechanisms conducting this VIF evaluation provides additional confidence in the model's interpretability and ensures that feature importance rankings will reflect genuine predictive contributions rather than artifacts of correlated inputs.

Exploratory Data Analysis

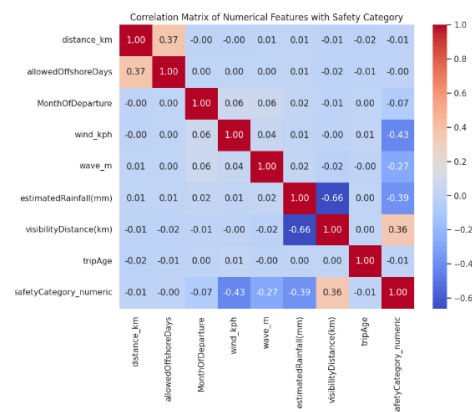
Distribution of safety category

The training dataset exhibits a notable class imbalance with approximately 10500 trips classified as 'safe' compared to roughly 5500 trips classified as 'unsafe'. This distribution indicates that safe trips represent approximately 66% of the dataset while unsafe trips constitute 34% which may require consideration of balancing techniques during model development. This imbalance suggests that the predictive model will need appropriate handling strategies to avoid bias toward the majority class.



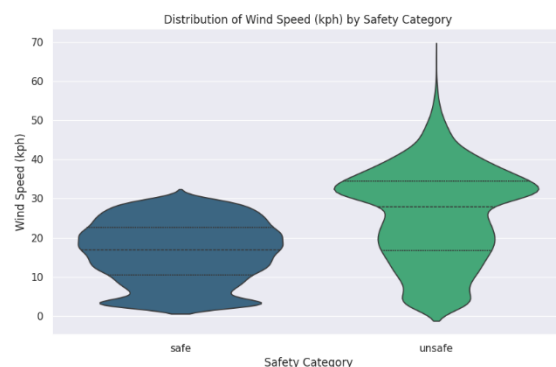
Correlation Matrix of Numerical Features with Safety Category

The correlation analysis reveals moderate negative relationships between the safety category and key environmental variables of wind speed (-0.43), estimated rainfall (-0.39), and wave height (-0.27) showing the strongest correlations. Visibility distance shows a positive correlation (0.36) with safety category suggesting that better visibility conditions are associated with safer trips.



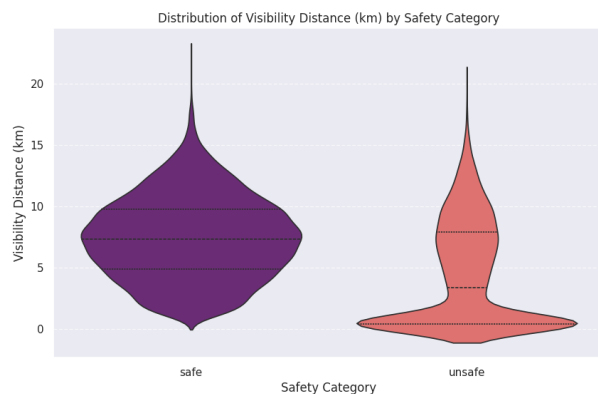
Distribution of Wind Speed (kph) by Safety Category

Unsafe trips are associated with distinctly higher wind speeds with the distribution centered around 27-35 kph and extending to approximately 70 kph, whereas safe trips demonstrate a more concentrated distribution around 17-23 kph. This clear differentiation establishes wind speed as a potentially strong predictor of trip safety with elevated wind speeds representing as a significant risk factor.



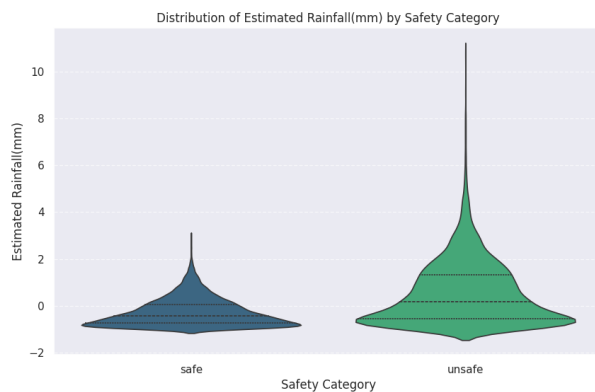
Distribution of Visibility Distance (km) by Safety Category

Safe trips are characterized by higher visibility distances with the distribution centered around 7-8 km, while unsafe trips show a bimodal distribution concentrated at lower ranges around 3-8 km. The presence of many unsafe trips at extremely low visibility highlights reduced visibility as a critical safety concern. The marked difference in median values between the two categories suggests visibility distance serves as a reliable discriminator for safety classification.



Distribution of Estimated Rainfall(mm) by Safety Category

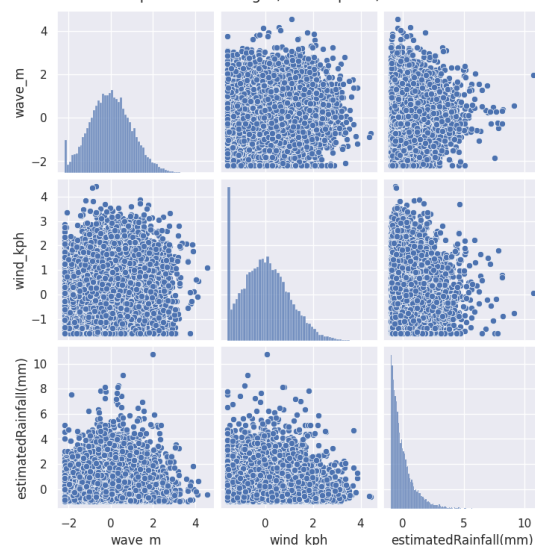
Unsafe trips experience significantly higher rainfall levels with the distribution extending from approximately -0.5 mm to 11 mm and centered around 1.5 mm compared to safe trips. The substantial separation between distributions indicates that precipitation is a meaningful safety indicator with higher rainfall levels strongly associated with hazardous conditions. The near zero rainfall observed during most safe trips suggests that favorable weather conditions with minimal precipitation are critical for safe offshore trips.



Relationship between Wave height, Wind speed and Estimated rainfall

This reveals weak to moderate positive relationships among the three environmental variables with wave height and wind speed showing the clearest association while estimated rainfall demonstrates more scattered patterns with both variables. The diagonal histograms indicate that wave height and wind speed follow approximately normal distributions whereas estimated rainfall exhibits a strong right skewed distribution with most values concentrated near zero. The lack of strong linear relationships suggests these environmental factors operate independently though their combined effects likely contribute to overall trip safety in a multivariate manner.

Pairwise Relationships of Wave Height, Wind Speed, and Estimated Rainfall



Data Preprocessing

Feature Scaling: Continuous variables were scaled using standardization techniques. Although tree-based models do not strictly require scaling, this step ensured consistency across all the models.

Encoding of Categorical Variables: Categorical predictors were encoded using appropriate techniques (One hot encoding). Encoding preserved category level information while making the data suitable for machine learning algorithms.

Model Development and Evaluation

Multiple machine learning algorithms were evaluated to identify the most suitable classifier for predicting offshore fishing trip safety including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting and XGBoost. Each model was trained on 16,000 samples and evaluated on 4,000 test samples with the safety category encoded as a binary outcome (0 = unsafe, 1 = safe).

Initial model Performance

Although Decision Tree, Random Forest and Gradient Boosting achieved perfect accuracy and F1 scores on the test set, that showed unusually high performance suggested potential overfitting. Therefore, hyperparameter tuning was carried out to improve generalization and ensure more reliable performance on unseen data.

Hyper parameter tuning

After hyperparameter tuning, the model was reevaluated on both the train and test datasets and the resulting performance metrics are reported below to provide a more realistic and reliable assessment of the model's generalization ability.

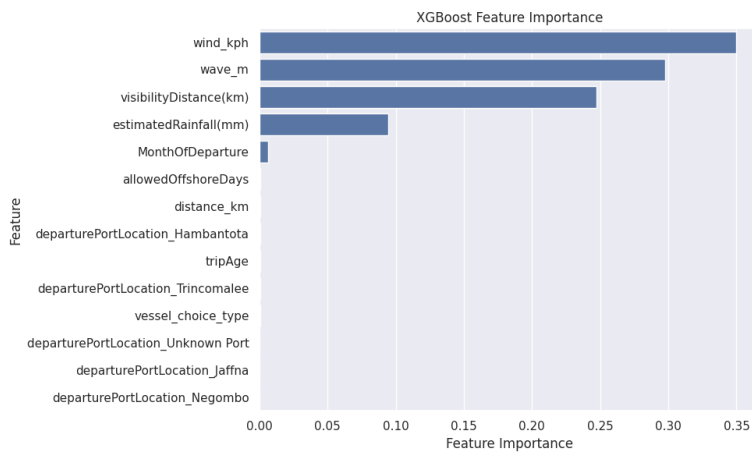
Model	Train Accuracy	Train F1 Score	Test Accuracy	Test F1 Score
Random Forest	1.00000	1.00000	1.00000	1.00000
Decision Tree	1.00000	1.00000	1.00000	1.00000
Gradient Boosting	1.00000	1.00000	1.00000	1.00000
XGBoost	0.99781	0.99834	0.99800	0.99847
Support Vector Machine	0.97144	0.97815	0.96350	0.97202
Logistic Regression	0.82806	0.87297	0.81275	0.86086

Final Model Selection

Despite hyperparameter tuning, Decision Tree, Random Forest and Gradient Boosting continued to exhibit perfect performance, confirming persistent overfitting issues. Hence, XGBoost was selected as the final predictive model due to its strong yet realistic performance achieving 99.80% accuracy and a 0.9985 F1 score which indicates effective learning without evidence of severe overfitting.

The model effectively handled class imbalance using built in weighting mechanisms and advanced regularization while its ability to capture complex nonlinear feature interactions without manual feature engineering made XGBoost more suitable for real world deployment than Random Forest and Gradient Boosting which exhibited overfitting in this application.

XGBoost Feature Importance Analysis



The feature importance analysis reveals that environmental variables overwhelmingly dominate safety predictions with wind speed (35.0%), wave height (29.8%), visibility distance (24.7%) and estimated rainfall (9.4%) collectively accounting for 99.0% of the model's predictive power. Wind speed emerges as the primary predictor while wave height and visibility contribute substantially to safety classification.

In contrast, operational factors such as departure port location, vessel type, trip distance and allowed offshore days demonstrate minimal predictive importance indicating that safety outcomes are predominantly driven by weather and sea conditions rather than trip planning variables. This distribution provides clear operational guidance that real time monitoring of environmental conditions should form the core of marine safety warning systems with operational variables serving as secondary considerations in risk assessment frameworks.

Appendix Including Python Code and Technical Details

Python Source Code: <https://colab.research.google.com/drive/19WiY7a1bckU19H2P5A-Vt3Yb4WphJMc9?usp>

Dataset link: <https://www.kaggle.com/datasets/dataanalystswrs/sri-lankan-offshore-trip-data-202225/data>

Tools used:

- As the coding environment we've used **Google Colaboratory**: A hosted Jupyter Notebook service that requires no setup to use and provides free access to computing resources, including GPUs and TPUs.
- Microsoft Word for documentation

References:

Report on cost-benefit analysis of the Monitoring, Control and Surveillance (MCS) System and tools developed by Sri Lanka. (n.d.). Retrieved February 2, 2026, from <https://fisheriesdept.gov.lk/wp-content/uploads/2025/02/Report-on-cost-%E2%80%93-benefit-analysis-of-the-monitoring-control-and-surveillanceMCS-system-and-tools-developed-by-Sri-Lanka.pdf>

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